Type MS · Calculating the range

Answer 1 of 6 · Question 3 part a · 1 mark

2025 JUN · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

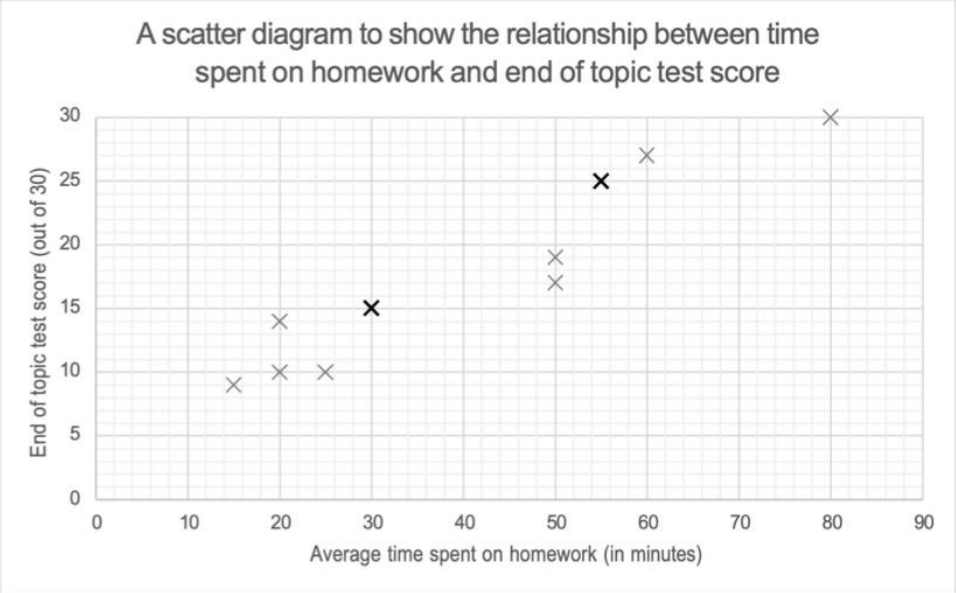
Question Number	Answer	Mark
3(a)	**One** mark for correct calculation of the range. Calculation: 30-9 • 21 Accept any other appropriate response.	**(1)**

Question Number	Answer	Mark
3(b)	**One** mark for calculation of number of students spending more than 30 minutes on homework. **One** mark for correct calculation of students spending more than 30 minutes on homework as a fraction of all students in the lowest form. • Total students = 10 and number of students spending more than 30 minutes = 5 • $\frac{5}{10}$ simplified $\frac{5}{5}$ and $\frac{10}{5} = \frac{1}{2}$ Accept any other appropriate response.	**(2)**

Type MS · Calculating the range

Answer 1 of 6 · Question 3 part a · 1 mark

2025 JUN · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

Question Number	Answer	Mark
3(c)	**One** mark for correct plotting of participant C. **One** mark for correct plotting of participant E. **One** mark for appropriate label and scale for x-axis. For example:  Accept any other appropriate response.	**(3)**

Type MS · Calculating the range

Answer 1 of 6 · Question 3 part a · 1 mark

2025 JUN · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

Type MS · Calculating the range

Answer 2 of 6 · Question 2 part c · 1 mark

2020 NOV · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

Question number	Answer	Mark
2(a)	**One** mark for calculating the mean. **One** mark for answer to two decimal places. • Calculation: 16+28+27+34+35+26+21= 187/7 = 26.714285 • 26.71 Reject all other answers.	(2)

Type MS · Calculating the range

Answer 2 of 6 · Question 2 part c · 1 mark

2020 NOV · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Mark
2(b)	**One** mark for correct rank ordering. **One** mark for correct answer. • Rank ordering: 18 24 29 30 31 35 38 • 30 Accept any other appropriate response.	(2)

Question number	Answer	Mark
2(c)	**One** mark for correct calculation. Calculation: 38 - 16 • 22 Accept any other appropriate response.	(1)

Type MS · Calculating the range

Answer 3 of 6 · Question 3 part c · 1 mark

2021 NOV · Paper 02 · find part c in this mark scheme extract

Question number	Answer	Mark
3(a)	**One** mark for calculating the mean. Calculation: 458+211+641+189=1499/4 • 374.75 Accept any other appropriate response.	(1)

Question number	Answer	Mark
3(b)	**One** mark for a correct calculation. 458+211+641+189+931+439+838+271 • 3978 Reject all other answers.	(1)

Question number	Answer	Mark
3(c)	**One** mark for correct calculation. Calculation: 931-271 • 660 Accept any other appropriate response.	(1)

Type MS · Calculating the range

Answer 4 of 6 · Question 4 part d · 1 mark

2022 JUN · Paper 02 · extract 1 of 2 · find part d in this mark scheme extract

Question number	Answer	Mark
4(a)	**One** mark for understanding of open-ended questions. **One** mark for application to the scenario. • An open-ended question allows respondents to answer freely using their own words and thoughts (1) so Jamie could gather detailed information about why the participants felt they did or did not obey authority figures (1). Accept any other appropriate response.	(2)

Question number	Answer	Mark
4(b)	**One** mark for stating one ethical issue that should be considered in the investigation. • All participants should have the right to withdraw from an investigation (1). Accept any other appropriate response.	(1)

Question number	Answer	Mark
4(c)	**One** mark for correct rank ordering. **One** mark for correct answer. • Rank ordering: 13 14 14 15 16 18 18 19 • (15+16)/2 = 15.5 Accept any other appropriate response.	(2)

Question number	Answer	Mark
4(d)	**One** mark for correct calculation. Calculation: 19 - 13 • 6 Accept any other appropriate response.	(1)

Type MS · Calculating the range

Answer 4 of 6 · Question 4 part d · 1 mark

2022 JUN · Paper 02 · extract 2 of 2 · find part d in this mark scheme extract

Type MS · Calculating the range

Answer 5 of 6 · Question 1 part d · 1 mark

2023 JUN · Paper 02 · extract 1 of 3 · find part d in this mark scheme extract

Question number	Answer	Mark
1(a)	B – gender **The only correct answer is B** A is incorrect as this is not the independent variable. C is incorrect as this is not the independent variable. D is incorrect as this is not the independent variable.	(1)

Question number	Answer	Mark
1(b)	**One** mark for each correct ethical issue given (maximum **two** marks) For example: • Informed consent can be gained by explaining the study to parents when asking if the children can take part (1). • Gary could use pseudonyms for the children to make sure none of them could be identified from his research (1). Accept any other appropriate response.	(2)

Question number	Answer	Mark
1(c)	**One** mark for calculation of total number of participants. **One** mark for calculation of percentage. • 50+50 = 100 • (25/100)*100 = 25% Accept any other appropriate response.	(2)

Question number	Answer	Mark
1(d)	**One** mark for correct calculation. Calculation: 32-24 • 8 Accept any other appropriate response.	(1)

Type MS · Calculating the range

Answer 5 of 6 · Question 1 part d · 1 mark

2023 JUN · Paper 02 · extract 2 of 3 · find part d in this mark scheme extract

Question number	Answer	Mark
1(e)	**One** mark for correct calculation of number of children.	(2)
	One mark for correct answer. Calculation: • 32+24= 56 Calculation: 56:100 simplified using a common factor • 14:25 Accept any other appropriate response.	

Question number	Answer	Mark
1(f)	**One** mark for conclusion made. **One** mark for justification of conclusion through analysis of data. • At six years old, girls can conserve number more accurately than boys (1) as more than twice as many girls than boys identified that the rows contained the same number of buttons (1). Accept any other appropriate response.	(2)

Type MS · Calculating the range

Answer 5 of 6 · Question 1 part d · 1 mark

2023 JUN · Paper 02 · extract 3 of 3 · find part d in this mark scheme extract

Type MS · Calculating the range

Answer 6 of 6 · Question 3 part a · 1 mark

2024 JUN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

Question number	Answer	Mark
3(a)	**One** mark for correct calculation. Calculation: 65 - 18 • 47 Accept any other appropriate response.	(1)

Question number	Answer	Mark
3(b)	**One** mark for correct calculation of remaining participants. **One** mark for correct answer. Calculation: (180/3)*2= • 120 participants Calculation: 60:120 simplified using a common factor • 1:2 Accept any other appropriate response.	(2)

Question number	Answer	Mark
3(c)	**One** mark for conclusion made. **One** mark for justification of conclusion through analysis of data. • The more hours of music a person listens to then the more likely they are to feel happy (1) as there is a positive correlation between hours listening to music and self-rated happiness scores out of 10 (1). Accept any other appropriate response.	(2)

Type MS · Calculating the range

Answer 6 of 6 · Question 3 part a · 1 mark

2024 JUN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract